Using matching for CATE estimation with post-ob&rct data

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OB - overlap

- Overlap condition influences the performance of estimators (Busso, DiNardo, and McCrary 2014)
- Theoretically, Overlap assumption is a must for the identification of ATE and Strict Overlap is needed for statistical analysis in many papers. (e.g. Causal RF)
- (Crump et al. 2009) [ATE] Characterize(by ps score) optimal subsamples for which the average treatment effect can be estimated most precisely.

we also consider the problem without imposing this restriction. Formally, we choose an estimand $\tau_{S,\mathbb{A}}$ by choosing the set $\mathbb{A} \subset \mathbb{X}$ that minimizes

$$V_{\mathbb{A}}^{\text{eff}} = \frac{1}{\{E(1_{X\in\mathbb{A}})\}^2} E\left[1_{X\in\mathbb{A}}\left\{\frac{\sigma_1^2(X)}{e(X)} + \frac{\sigma_0^2(X)}{1-e(X)}\right\}\right].$$

Defining $q(\mathbb{A}) = \text{pr}(X \in \mathbb{A}) = E(1_{X\in\mathbb{A}})$, we can write the objective function as

$$V_{\mathbb{A}}^{\text{eff}} = \frac{1}{q(\mathbb{A})} E\left\{\frac{\sigma_1^2(X)}{e(X)} + \frac{\sigma_0^2(X)}{1-e(X)} \middle| X \in \mathbb{A}\right\}.$$

We seek $\mathbb{A} = \mathbb{A}^*$, which minimizes $V_{\mathbb{A}}^{\text{eff}}$ among all closed subsets $\mathbb{A} \subset \mathbb{X}$.

- (Li, Thomas, and Li 2019) [ATE] Overlap Weighting

OB - overlap

- why poor overlap?
  -
- poor overlap, then?
- when does that matter

OB - overlap & balance(in high-dimensional HTE estimation)

- overlap and balance

$$xx$$

- high-dimensional: (1) many covariates; (2) complex interactions among covariates in outcome model or propensity model
- But strict overlap is very 'strict' in high-dimensional setting. (D'Amour et al. 2021)[1]
- From data-generating view, if we have high-interaction confounding, then poor high-dimension overlap [?]

[1] Strict overlap restricts global discrepancies between the covariate distributions in the treated and control populations. These restrictions become more restrictive as the dimension grows large.

CRF - high-dimensional/heterogeneity

- Bias of estimation(Wager and Athey 2018)

$$|\mathbb{E}[\hat{\tau}(x)] - \tau(x)| \lesssim 2Md\left(\frac{\eta s}{2k-1}\right)^{-\frac{1}{2}\frac{\log((1-\alpha)^{-1})}{\log(\alpha^{-1})}\frac{\pi}{d}} = \mathcal{O}\left(s^{-\frac{1}{2}\frac{\log((1-\alpha)^{-1})}{\log(\alpha^{-1})}\frac{\pi}{d}}\right)$$

  - bounded covariates $X_1, \ldots X_s \in [0,1]^d$
  - M-bounded and Lipschitz continuous conditional mean functions
  - strict overlap: for some $\eta > 0$ and all $x \in [0,1]^d$, $\eta < e(x) < 1 - \eta$.
  - A tree predictor grown by recursive partitioning is α-regular
  - (for consistency) leaves of the trees become small in 'all' dimensions of the feature space as n gets large. At each step, each variable is selected w.p. at least π/d
- Tree beats NN under settings of low-dimensional signal structure[2]
  - signal dimensions q and ambient dimensions d

		mean-squared error			coverage		
q	d	CF	10-NN	100-NN	CF	10-NN	100-NN
2	6	0.04 (0)	0.24 (0)	0.13 (0)	0.92 (1)	0.92 (0)	0.59 (1)
4	6	0.06 (0)	0.22 (0)	0.07 (0)	0.87 (1)	0.93 (0)	0.72 (1)
6	6	0.08 (0)	0.22 (0)	0.05 (0)	0.75 (1)	0.93 (0)	0.78 (1)
2	12	0.04 (0)	0.38 (0)	0.34 (0)	0.86 (1)	0.88 (0)	0.45 (0)
4	12	0.08 (0)	0.30 (0)	0.18 (0)	0.76 (1)	0.90 (0)	0.51 (1)
6	12	0.12 (0)	0.26 (0)	0.12 (0)	0.59 (1)	0.91 (0)	0.59 (1)

- More[3]: could match/balance based weighting be used in CRF?

[2] a data-driven way to determine which nearby observations receive more weight

[3] "An important challenge for future work is to design splitting rules that can automatically choose which characteristic of the training data to split on."

OB - define/calculate balance

Need a metric to describe the distance of two distributions.

- one-dimensional norm(e,g. overlap region of propensity score[4]); mean-matching/pair-matching (Rubin 1973a)
- Mahalanobis distance[5] (which also contain terms that are functions of the covariates like power/interactions)
- p-values form KS test and paired t-tests (general because the same scale) (Diamond and Sekhon 2013)
- combine ps-score and original X and ..(better firstly orthogonalize by regressing)

[4] for ps score, better to use the linear predictor because it is note close to 0 or 1 (Diamond and Sekhon 2013)

[5] MD does not perform well when covariates have nonellipsoidal distributions?

OB - matching

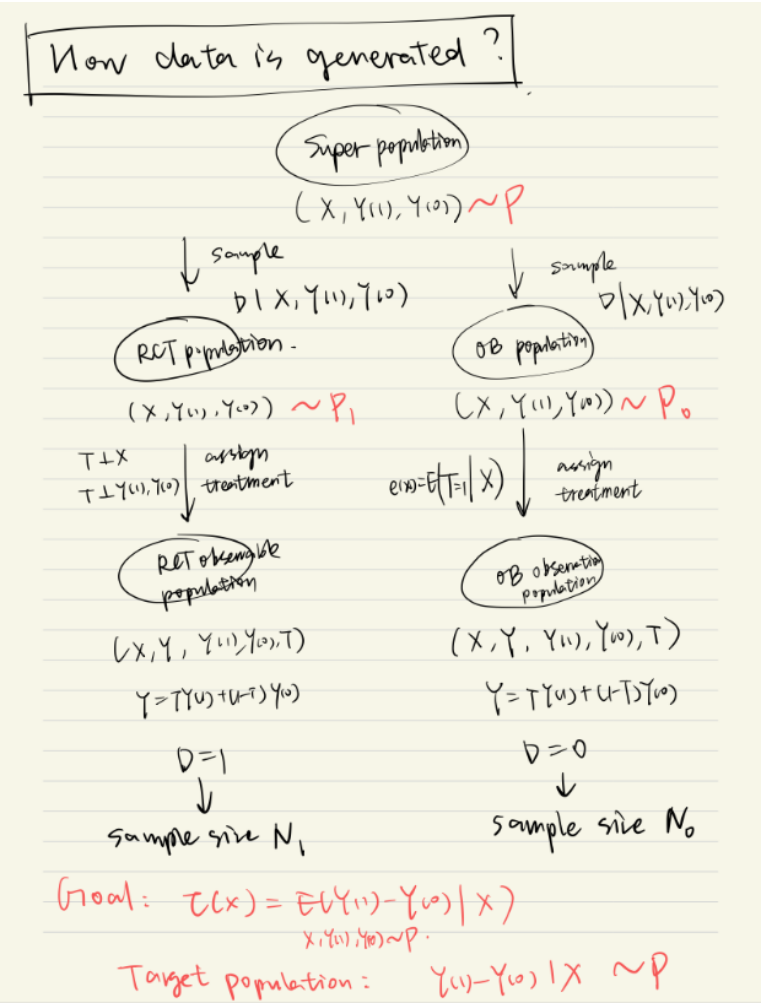
- (Rubin 1973b) [OB, ATT, one-dimension X, homogeneous/parallel effect]. Use percent reduction in bias as a metric. Matching and regression can reduce bias in observational studies.
- (Diamond and Sekhon 2013) The GenMatch algorithm searches a range of distance metrics to find the particular measure that optimizes post-matching covariate balance. iteratively checking the specification of the propensity score model.

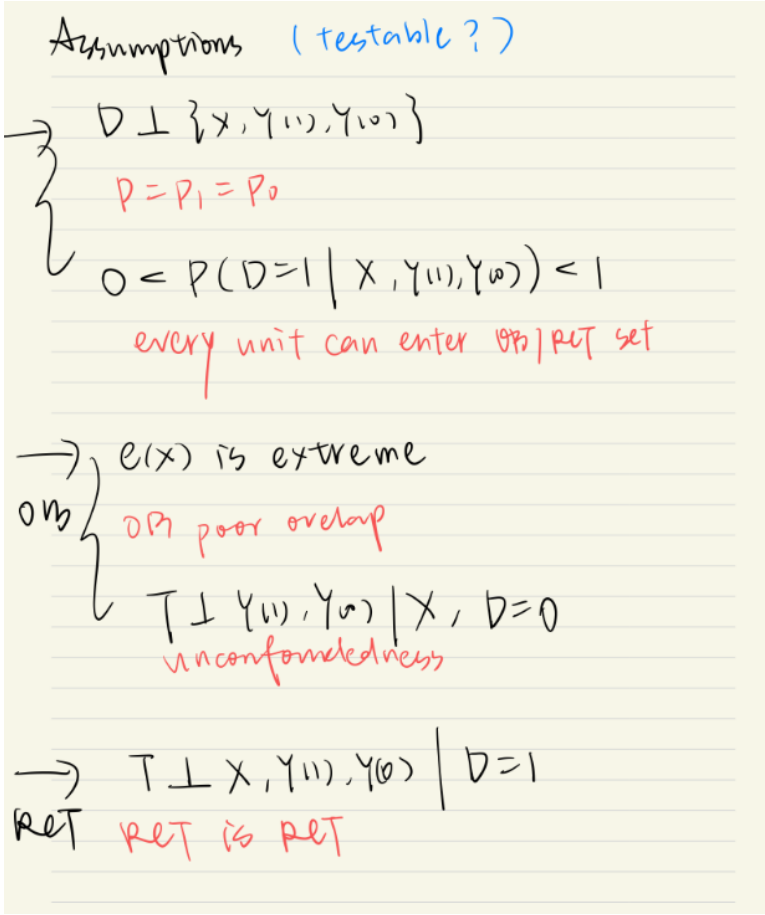
combine rct and ob - why?

- RCT+EC
  - RCT so small
  - RCT may only cover some specific patient(X support region)
- OB+RCT
  - OB with poor overlap(in what sense?)
  - OB with unmeasured confounding

combine rct and ob - a data generating view

In super population framework, both RCT and OB are samples from a stochastic distribution. (not finite population and design-based experiment framework)





combine rct and ob

- (Liu et al. 2025) Introduce a double-indexed notation for potential outcomes to define causal estimands transparently and clarify distinct sources of implicit bias. A_i treatment status; Z_i data source.

$$Y_i = Y_i^1(1)A_iZ_i + Y_i^1(0)(1-A_i)Z_i + Y_i^0(1)A_i(1-Z_i) + Y_i^0(0)(1-A_i)(1-Z_i)$$

$$b(X_i) = E(Y_i \mid Z_i = 1, A_i = 0, X_i) - E(Y_i \mid Z_i = 0, A_i = 0, X_i)$$

- (Kim et al. 2025) Define weak overlap; covariate shift between sources.

Simulations settings

Under what kind of settings would our methods be better than other? N obs data, M rct data; d covariates, within which c are confounders and $d - c$ are prognsotic covariates(assume no instrumental variables)

- N/M
- $e(x)$, overlap, well-specified or not
- $\mu_0(x), \mu_1(x)$, the response surfaces and CATE $\tau(x)$
- c/d, how many irrelevant covariates
- high-dimensional or not (d/N or c/N, complexity of models)

Simulations settings - how to simulate X

(D'Amour et al. 2021) Covariate sequence X is a stochastic process $(X_{(k)})_{(k>0)}$, for any single problem, the investigator selects a finite set of covariates $X_{1:p}$ from the finite pool of covariates $(X_{(k)})_{(k>0)}$.

Simulations settings - linearity of outcome model

If the true outcome model is linear, $\mu_{(w)}(x) = x\beta(w)$, then

- the difference of them, i.e. CATE, is linear in X
- fitting one model for each z is equivalent to adding an interaction term Xz
- achieving sample balance[6] just involves balancing the X_i
- If the response surface is nonlinear, all estimates are general biased even if $\bar{x}_1 = \bar{x}_2$. (Rubin 1973b) [ATT, one-dimension]

[6] $\frac{1}{n}\sum_{i=1}^{n}\frac{Z_i\phi_j(X_i)}{e(X_i)} = \frac{1}{n}\sum_{i=1}^{n}\frac{(1-Z_i)\phi_j(X_i)}{1-e(X_i)} = \frac{1}{n}\sum_{i=1}^{n}\phi_j(X_i)$

Some thoughts and questions

- why does overlap influence the performance of estimators in math[7]
- What is target population? After matching/weighting, would that actually change? (can visualize X)
- After matching/weighting, is the balance better?
- metrics of CATE estimation for simulation? [8]
- data-augmenting then modeling v.s weighting in DL/Tree models (would it be different if we double one sample or weight it by two in model?)
- poor overlap in high-order-interaction-terms sense? hard to grow a tree? [how much overlap, i.e. how strong tails of $e(x)$]

[7] 'If the imbalance of the covariates between the two groups is large, the model-based results heavily relies on extrapolation in the region with little overlap, which is sensitive to the model specification assumption.'

[8] (1) $\frac{1}{n}\sum_{i=1}^{n}(\hat{\tau}_i - \tau_i)^2$; (2) $\frac{1}{n}\sum_{i=1}^{n}\hat{\tau}(x_i) - \frac{1}{n}\sum_{i=1}^{n}\tau(x_i)$; (3) $E(\hat{\beta} - \beta)^2$

Summary

- Goal: better estimate CATE
  - data: large-scale Post-OB data and small-scale RCT data
  - problem: poor overlap and severe unbalance
- Method: matching-based
  - data augmentation
  - ITE-imputing loss
- Data source: simulation and real-world data
- Further
  - when would the methods work better(the choice of caliper for matching, the size of ob&rct, the degree of non-overlap, the use of previous algorithm or not, etc)
  - an end-to-end rather than step-wise algorithm
  - multi-treatment

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